Pokin' Tokens

Social Sports Prediction Platform

Project Overview & Technical Documentation

Version: 1.0.0

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Repository: github.com/ahelferthaus/2026-World-Cup

Status: Active Development

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1. Executive Summary

Pokin' Tokens is a cross-platform social sports prediction app built for the 2026 FIFA World Cup — the first World Cup hosted by three nations (USA, Canada, Mexico) and the first with 48 teams. The app targets high school students who want to predict match outcomes, challenge friends, and compete on school-based leaderboards using virtual tokens.

Key Differentiators:

- **School-based social competition** — Leaderboards segmented by school and state drive organic, word-of-mouth adoption within student communities
- **Virtual token economy** — No real money involved, making it school-appropriate and COPPA-friendly
- **AI-powered engagement** — Perplexity-powered research lab and Gemini-powered chatbot mascot ("Pokin' Pete")
- **Multi-sport extensibility** — Architecture supports expansion beyond soccer (NBA Playoffs already integrated)
- **Premium Gen-Z UI** — Dark theme with glassmorphism, animated gradients, and neon accents

Product Metrics (Target)

Metric	Target	Driver
DAU during World Cup	10,000+	Daily predictions + streak system
School penetration	500+ US high schools	School leaderboards + challenges
Session length	8+ minutes	Live game betting + AI chat
D7 retention	40%+	Streak rewards + daily challenges

2. Technology Stack

Frontend

Component	Technology	Version	Purpose
Framework	Flutter	3.7+	Cross-platform UI (iOS, Android, Web)
Language	Dart	3.7.0	Type-safe, AOT-compiled for mobile
State Management	Riverpod	3.0	Reactive state, dependency injection
Navigation	GoRouter	16.0	Declarative routing with auth guards
Design System	Material 3	Latest	Dark theme with custom color scheme
HTTP	http package	1.3	Cloud Function API calls

Backend

Component	Technology	Version	Purpose
Runtime	Node.js	22	Cloud Function execution
Language	TypeScript	5.7	Type-safe backend code
Functions	Firebase Functions v2	6.3	Serverless HTTP & event handlers
Admin SDK	Firebase Admin	13.0	Server-side Firestore & Auth access
HTTP Client	Axios	1.8	External API calls

Infrastructure

Service	Provider	Purpose
Database	Cloud Firestore	NoSQL document database
Authentication	Firebase Auth	Google, Apple, Email sign-in
Serverless Functions	Firebase Cloud Functions	API proxy, business logic, scheduled jobs

Sports Data	API-Football	Live fixtures, scores, standings
AI Research	Perplexity AI (sonar)	Real-time football research with citations
AI Chat	Google Gemini 2.0 Flash	Chatbot personality engine
Source Control	GitHub	Code repository and version control

3. Architecture & Infrastructure

System Architecture

```
┌──────────────────────────────────────────┐
│              Flutter App                 │
│  ┌──────────┐ ┌──────────┐ ┌──────────┐  │
│  │ iOS App  │ │ Android  │ │ Web App  │  │
│  │ (native) │ │ (native) │ │ (WASM)   │  │
│  └────┬─────┘ └────┬─────┘ └────┬─────┘  │
│       └────────────┴────────────┘        │
│              Riverpod State              │
│              GoRouter Navigation         │
└──────────────────┬───────────────────────┘
                   │ HTTPS
┌──────────────────┴───────────────────────┐
│         Firebase Cloud Functions         │
│  ┌──────────────────────────────────┐    │
│  │  HTTP Endpoints (7)              │    │
│  │  matchesToday | standings | predictions│ │
│  │  leaderboard  | research  | chat       │ │
│  └──────────────────────────────────┘    │
│  ┌──────────────────────────────────┐    │
│  │  Triggers                        │    │
│  │  onUserCreated | resolveMatchPredictions│ │
│  └──────────────────────────────────┘    │
└──────┬──────────┬──────────┬─────────────┘
       │          │          │
┌──────┴──┐  ┌────┴─────┐  ┌─┴──────────┐
│Firestore│  │API-Football│ │AI APIs     │
│         │  │          │  │Perplexity  │
│users    │  │fixtures  │  │Gemini      │
│predicts │  │standings │  │            │
│cache    │  │          │  │            │
└─────────┘  └──────────┘  └────────────┘
```

Data Flow: Making a Prediction

1. User selects a match and picks an outcome (e.g., "USA wins")
2. Flutter sends POST to `createPrediction` Cloud Function with auth token
3. Function verifies auth, validates wager against user balance
4. Function creates prediction doc in Firestore, decrements user tokens atomically

5. Every 5 minutes, `resolveMatchPredictions` checks API-Football for completed matches
6. For each completed match, function evaluates pending predictions
7. Winners receive `wager × multiplier` tokens via `FieldValue.increment()`

Caching Strategy

API-Football has rate limits, so all sports data is cached in Firestore:

Cache	TTL	Rationale
Match fixtures	2 minutes	Near-real-time for live scores
Standings	30 minutes	Changes only after match completion

4. Complete Feature Set

4.1 Prediction Markets

Market Type	Description	Payout
Match Winner (1X2)	Home, Draw, or Away	2.0x wager
Exact Score	Predict final score (e.g., 2-1)	5.0x wager
Over/Under 2.5	Total goals above or below 2.5	Odds-based
Both Teams to Score	Yes or No	Odds-based
Player Props	Anytime/First Goalscorer, Assists, Shots	Odds-based
ALL IN	Risk entire token balance on one pick	Standard multiplier

4.2 Social & Competitive Features

- **P2P Challenges** — Create head-to-head bets in 3 categories (Sports, Custom, Party Game)
- **Social Feed** — Venmo-style activity stream showing challenges, wins, and losses
- **Head-to-Head Records** — Track win/loss record against each opponent
- **Party Mode** — Multiplayer lobby for group challenges at watch parties
- **4-Tab Leaderboard** — Global, My School, By School aggregate, By State aggregate

4.3 Live Game Experience

- Real-time scoreboard with team logos and match minute
- Quick bet chips (Next Goal, No Goals, Quick Goal) during live matches
- VS tally bar showing running token count against friends
- Game event feed with goal announcements, cards, and substitutions
- Instant bet resolution on match events

4.4 Progression & Retention

- **Daily Challenges** — 4 challenges per day with token rewards (5-15 tokens each)
- **Weekly Challenges** — 3 challenges per week (25-100 tokens each)
- **Streak Tracking** — Visual streak counter with milestone rewards at 3/5/7/14/30/60 days
- **Token Store** — 4 purchasable token packs ($0.99 to $14.99)

4.5 Content & Discovery

- **Tournament Bracket** — Full knockout visualization (R16 through Final)
- **Multi-Sport Hub** — Expandable sport cards (World Cup + NBA Playoffs)
- **Teams Encyclopedia** — Squads, FIFA rankings, coaches, Golden Boot tracker
- **Player Detail Sheets** — Stats, position, club, age, international caps
- **News Feed** — Categorized sports articles with source attribution

4.6 AI Features

- **Research Lab** — Perplexity AI-powered football research with source citations
- **Pokin' Pete Chatbot** — Gemini-powered AI mascot with soccer personality

4.7 Screen Inventory (20+ screens)

Screen	Route	Nav Type
Home (Matches + Standings)	`/home`	Bottom Nav
Predict (Winner + BTTS + Props)	`/predict`	Bottom Nav
Leaderboard (4 tabs)	`/leaderboard`	Bottom Nav
Profile + Edit + History	`/profile`	Bottom Nav
Challenges Hub (5 tabs)	`/propbets`	Full Screen
Live Game	`/live/:matchId`	Full Screen
Tournament Bracket	`/bracket`	Full Screen
Teams (Teams + Players + Golden Boot)	`/teams`	Full Screen

Multi-Sport Hub	`/sports`	Full Screen
NBA Playoffs	`/nba`	Full Screen
News Feed	`/news`	Full Screen
Token Store	`/store`	Full Screen
Streaks & Challenges	`/streaks`	Full Screen
Research Lab	`/research`	Full Screen
Pokin' Pete Chat	`/chat`	Full Screen
Login / Register	`/login`	Auth Flow

5. Platform Strategy: iOS App vs Web App

How Flutter Enables Multi-Platform

Flutter compiles from a single Dart codebase to three distinct targets:

Platform	Build Command	Output	Distribution
iOS	`flutter build ios`	Native iOS binary (ARM64)	Apple App Store / TestFlight
Android	`flutter build appbundle`	Android App Bundle	Google Play Store
Web	`flutter build web`	HTML/JS/WASM	Firebase Hosting / any CDN

iOS App Setup

The project is fully configured for iOS deployment:

- **Xcode project** — `ios/Runner.xcodeproj` and `Runner.xcworkspace` (CocoaPods)
- **Bundle info** — `Info.plist` with display name "Pokin' Tokens", portrait + landscape orientations
- **Firebase** — `GoogleService-Info.plist` generated via FlutterFire CLI
- **Auth** — Apple Sign-In configured (required for App Store apps with social login)
- **Build** — Requires macOS with Xcode; archive and upload to App Store Connect

Web App Setup

The web build is a Progressive Web App (PWA) with:

- **index.html** — Flutter bootstrap loader, responsive viewport, PWA meta tags
- **manifest.json** — PWA manifest for "Add to Home Screen" functionality
- **Icons** — Favicon and touch icons for mobile bookmarks
- **Responsive layout** — Adapts from mobile (bottom nav) to tablet (side rail) to desktop (extended rail)

Shared vs Platform-Specific Code

> **100% shared Dart code.** There is no platform-specific Dart code. All UI, business logic, navigation, and state management is shared across iOS, Android, and Web. The only platform-specific files are native configuration (Info.plist, AndroidManifest.xml, index.html) which Flutter manages automatically.

Deployment Strategy

Phase	Platform	Rationale
Phase 1 (MVP)	Web (Firebase Hosting)	Fastest to deploy, no app store review, shareable links for school adoption
Phase 2 (Pre-WC)	iOS (App Store)	Premium experience for iPhone users (dominant in US teen market)
Phase 3 (Launch)	Android (Play Store)	Expand reach for broader adoption

6. AI Integration

Research Lab (Perplexity AI)

Attribute	Value
Model	Perplexity `sonar`
Proxy	Firebase Cloud Function (`/research`)
System prompt	Concise football research, <400 words, markdown formatted
Recency filter	Last month
Output	Markdown content + up to 4 source citation URLs
Use case	Users research players, teams, and stats before making predictions

Pokin' Pete Chatbot (Google Gemini)

Attribute	Value
Model	Gemini 2.0 Flash
Proxy	Firebase Cloud Function (`/chat`)
Personality	Witty sports commentator, enthusiastic, school-appropriate
Temperature	0.9 (creative, varied responses)
Max tokens	1,024 per response
Guardrails	No profanity, no inappropriate content, steers to soccer topics
Features	Multi-turn conversation, starter prompts, markdown rendering

AI Cost Considerations

Both AI features are proxied through Cloud Functions, enabling:

- API key security (keys never exposed to client)

- Rate limiting at the function level
- Response caching potential (common queries)
- Usage monitoring via Firebase function logs
- Easy model swapping without app updates

7. Security Model

Defense in Depth

Client Layer

- Firebase Auth tokens required for all API calls
- No API keys stored in client code
- Auth state managed by Riverpod with automatic refresh

API Layer

- Cloud Functions verify Bearer tokens via Firebase Admin SDK
- All external API keys stored as Firebase Secrets (not environment variables)
- CORS enabled with domain restrictions

Database Layer

- **Users:** Read own doc only; can only update `displayName` (string, 1-20 chars)
- **Predictions:** Read own only; all writes are server-side via Admin SDK
- **Cache:** Read-only for authenticated users
- **Default:** Deny all unmatched paths
- **Token writes:** Server-side only via `FieldValue.increment()` — users cannot modify their own balance

Business Logic

- Wager validation: server checks balance before accepting prediction
- Atomic token operations via Firestore transactions
- Prediction resolution runs server-side on a schedule (no client trust)

8. Business Model & Monetization

Revenue Streams

8.1 Virtual Token Purchases (Primary)

Package	Tokens	Price	$/Token
Starter Pack	100	$0.99	$0.0099
Fan Pack Popular	500	$3.99	$0.0080
Baller Pack	1,200	$7.99	$0.0067
Legend Pack	3,000	$14.99	$0.0050

Note: Tokens are virtual currency with no cash-out value, keeping the app compliant with gambling regulations. This is a prediction game, not gambling.

8.2 Premium Features (Future)

- **Pokin' Tokens Pro** — Monthly subscription for advanced analytics, unlimited AI research, custom avatars, and ad-free experience
- **Sponsored Challenges** — Brand-sponsored prediction challenges with real prize pools (gift cards, merch)
- **School Tournaments** — Premium inter-school tournament brackets with entry fees

8.3 Advertising (Future)

- Native ad placements in news feed
- Sponsored sports content and branded prediction cards
- Targeted to 14-18 demographic (COPPA-compliant ad networks)

Unit Economics Model

Metric	Assumption

Free users	85% of total (tokens from signup + streaks)
Paying users	15% conversion
Avg purchase	$3.99/user (Fan Pack)
Purchases/user/month	1.5x during World Cup
ARPU (paying)	~$6/month during tournament

Growth Strategy

1. **School Network Effect** — One student shares with friends → school leaderboard drives competition → word-of-mouth within school → cross-school rivalry drives expansion
2. **World Cup Timing** — Launch aligned with June 2026 tournament start for maximum organic interest
3. **Web-First Launch** — Shareable links (no app store friction) for fastest school penetration
4. **Content Loop** — AI Research + Pete chatbot keep users engaged between matches
5. **Streak Mechanics** — Daily/weekly challenges + streak rewards drive habitual return behavior

Competitive Landscape

Competitor	Differentiator
FanDuel / DraftKings	Real money gambling (18+); Pokin' Tokens is virtual & teen-appropriate
ESPN Bracket Challenge	One-time bracket only; PT has daily predictions + live betting + social
Sorare / Footium	NFT/crypto-based; PT is simple token system with no blockchain
FIFA Official App	No prediction game; PT fills the fan engagement gap

9. Roadmap & Growth Strategy

Phase 1: Pre-Tournament (Now – May 2026)

- Complete Firebase project setup and deploy Cloud Functions
- Connect live API-Football data (replace demo mode for production)
- Implement Stripe/Apple Pay for token purchases
- Deploy web app to Firebase Hosting
- Beta test with 5-10 schools
- AI avatar generation for profiles

Phase 2: Tournament Launch (June 2026)

- Submit to iOS App Store and Google Play
- Launch marketing push targeting high school sports communities
- Enable push notifications for live matches and challenge responses
- Real-time Firestore listeners for live game updates
- Direct messaging between challengers

Phase 3: Post-Tournament Expansion (August 2026+)

- Expand multi-sport hub: NFL, Champions League, Premier League, MLS
- Introduce Pokin' Tokens Pro subscription
- School ambassador program for organic growth
- Branded challenge partnerships with sponsors
- Fantasy-style season-long competition mode

Technical Debt & Infrastructure

- CI/CD pipeline (GitHub Actions for build + deploy)
- Automated testing (unit + widget + integration)
- Analytics integration (Firebase Analytics, Crashlytics)

- Performance monitoring and error tracking
- CDN for team logos and player images

10. Codebase Statistics

Metric	Count
Total source files	115
Total lines of code	~16,000
Dart files (frontend)	~85
TypeScript files (backend)	~15
Feature modules	18
Screens	20+
Reusable widgets	11
Cloud Functions	9 (7 HTTP + 1 auth trigger + 1 scheduled)
Firestore collections	4 + 2 cache collections
Composite indexes	4
Git commits	12
Dependencies (Flutter)	12 packages
Dependencies (Functions)	3 packages

Repository

Attribute	Value
URL	github.com/ahelferthaus/2026-World-Cup
Branch	`master`
Firebase Project	`pokin-tokens`
Flutter Package	`pokin_tokens`

Pokin' Tokens — Predict • Compete • Win

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